

FINAL PROJECT

QUANTITATIVE TRADING AND PRICE IMPACT MSC IN MATHEMATICS AND FINANCE 2025-2026

1. INTRODUCTION

The aim of the final project is for students to integrate the weekly homework into a single cohesive framework. This note outlines the various steps that comprise the framework, describes the baseline expectations for students, and suggests directions for extending the basic approach. Note that students should not attempt to implement all possible enhancements simultaneously. Instead, they should focus on executing two or three enhancements well and clearly communicating the impact of the added complexity.

Dates and Deadlines:

- (a) Presentations: May 15, 2026
- (b) Report Submission Deadline: May 15, 2026 (23:59)

2. PROJECT STEPS

2.1. Data Preparation. First, students should preprocess the data in a manner that facilitates efficient traversal for various fitting and simulation tasks. This involves splitting the data into in-sample and out-of-sample sets, and determining which files can be merged and which should remain separate to speed up computation.

The baseline requires fitting model parameters using in-sample data from 20 stocks over one month, with the model then applied to the next month's data for the same 20 stocks.

To expand, perform the study on a rolling basis over the entire universe of stocks. This enhancement is highly recommended to obtain more robust statistical metrics for the final simulation.

2.2. Impact Model Fitting. Second, students should fit multiple price impact models using the in-sample data from the public trading tape.

The baseline involves fitting the OW model along with one more advanced model, such as the AFS or a reduced-form model.

To expand, fit a non-parametric model and incorporate appropriate regularization. For example, one may fit a separate model parameter λ_i for each stock i and add a regularization term $||\lambda_i - \bar{\lambda}||^2$, where $\bar{\lambda}$ represents the parameter from the universal model. Note that this regularization requires

an additional sample, resulting in three samples: the training sample (in-sample), the testing sample (for regularization), and the validation sample (out-of-sample). Care should be taken to explain how the regularization hyperparameter was tuned.

2.3. Backtest Engine. Third, students should implement a backtest engine that can process generic trades rather than only optimal ones.

The baseline implements a version of Waelbroeck’s simulator, based on the fitted price impact model, which resets positions and impact states at the beginning of each day.

To expand, develop a backtest engine that tracks impact states and positions over multiple days. Be sure to account for the possibility of stock splits. This can be achieved by either hard-coding a list of affected stocks and dates or by detecting them when prices and volumes move in opposite directions. For example, prices may double while daily volumes halve, leaving the overall daily trade notional unchanged. Stocks undergoing splits should be excluded.

2.4. Synthetic Alphas. Fourth, students should implement synthetic look-ahead alphas for both the alpha level and decay.

The baseline employs the same synthetic alpha as used in the homework.

To expand, implement an additional synthetic alpha for overnight returns.

2.5. Optimal Trading Strategy. Fifth, students should develop trading strategies based on the alpha signals.

The baseline implements the optimal strategy for both the OW model and the fitted model.

To expand, implement the optimal trading strategy incorporating a synthetic overnight alpha. Simulate trading strategies both with and without the intraday alpha.

2.6. Performance Metrics. Sixth, students should compile a performance report for each strategy.

The baseline report should include the alpha’s correlation with returns, as well as key portfolio metrics such as expected daily P&L, Sharpe ratio, transaction costs, maximum daily drawdown, and maximum impact dislocation.

To expand, enhance the performance report to be as visually appealing and informative as possible.

2.7. Sensitivity Analysis and Stress Testing. Seventh, students should use their framework to explore questions about the trading strategies.

The baseline should outline the various degrees of freedom (e.g., choice of impact model, strength of the alpha signal) and illustrate how these affect strategy performance.

To expand, simulate scenarios to answer questions such as: “What if I use the wrong impact model?” “What if my signal is delayed by one minute?”

or “What if I am forced to liquidate my position midway through the day in a single block trade?”

3. FORMAT OF ASSESSMENT

The report is due on the day of the presentations. It should be polished and presentable, covering all the sections mentioned, though it does not need to be highly technical. For each section, if the procedure is standard, a simple example is sufficient; if the procedure is more advanced, additional outputs (such as images, sample paths, summary statistics, or any measure of the object’s size) are required. No code needs to be attached to the report, but using function signatures (detailing input/output) is encouraged to explain key functions. At the beginning of each section, students should indicate who contributed to that section and provide a brief explanation of what was done (whether it is a basic or advanced procedure). It is better to implement two well-executed enhancements beyond the basics than to implement several enhancements poorly.

The oral presentation is expected to be more technical. The main focus should be on what was done beyond the basics. During the presentation, each student will be asked questions about the specific parts of the course-work to which they contributed. Those who contributed to the coding should be prepared to discuss and describe their code.